Part IV

20 January 2024 02:03

1. Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Determine the properties of a partial ordering that the others lack.

a) $\{(0, 0), (1, 1), (2, 2), (3, 3)\}$

b) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 2), (3, 3)\}$

c) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 3)\}$

d) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 2), (2, 3), (3, 3)\}$

e) $\{(0, 0), (0, 1), (0, 2), (1, 0), (1, 1), (1, 2), (2, 0), (2, 2), (3, 3)\}$

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A partial ordering is a relation which is reflexive antisymmetric and transitive.

a)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

- $m_{ii} = 1 \Rightarrow$ reflexive
- $m_{ij} = m_{ji} = 0$ for $j \neq i \Rightarrow$ antisymetric & symmetric
- Since $M \odot M = M \Rightarrow M^{\odot n} = M \ \forall n \Rightarrow$ transitive

Thus, this relation is a partial ordering

b)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 1 & 1 \end{pmatrix} \end{array}$$

- $m_{ii} = 1 \Rightarrow$ reflexive
- $m_{23} = m_{32} = 1 \Rightarrow R$ is not antisymetric

This is not a partial ordering.

c)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

- $m_{ii} = 1 \Rightarrow$ reflexive
- $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji}$ for $i \neq j$ $\Rightarrow$ antisymmetric

$$M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} = M$$

Thus, $M^{\odot n} = M \ \forall n \Rightarrow M$ is transitive.

Since $M^{\odot n} \subseteq M$. $\Rightarrow R$ is a PO "partial order"

d)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$

- $m_{ii} = 1 \Rightarrow$ reflexive
- $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji}$ for $j \neq i$
  $\Rightarrow$ antisymmetric

- $M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = M$

Thus, $M^{[n]} = M \Rightarrow M^{[n]} \subseteq M \Rightarrow R$ is transitive. $\Rightarrow R$ is a PO

e)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 1 & 0 & \\ 1 & 1 & 0 & \\ & & & \\ & & & \end{pmatrix} \end{array}$$
← don't need to complet
$(1,0) \wedge (0,1) \in R$ with $0 \neq 1$
Thus, R is not antisymmetric

This is a PO.

2. Which of these relations on $\{0, 1, 2, 3\}$ are partial orderings? Determine the properties of a partial ordering that the others lack.

a) $\{(0, 0), (2, 2), (3, 3)\}$

b) $\{(0, 0), (1, 1), (2, 0), (2, 2), (2, 3), (3, 3)\}$

c) $\{(0, 0), (1, 1), (1, 2), (2, 2), (3, 1), (3, 3)\}$

d) $\{(0, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (2, 3), (3, 0), (3, 3)\}$

e) $\{(0, 0), (0, 1), (0, 2), (0, 3), (1, 0), (1, 1), (1, 2), (1, 3), (2, 0), (2, 2), (3, 3)\}$

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a)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix} \end{array}$$
$\Rightarrow$ not diagl $\Rightarrow R$ is not reflexive
$\Rightarrow R$ is not a partial ordering.

b)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \end{array}\begin{array}{c} \begin{array}{cccc} 0 & 1 & 2 & 3 \end{array} \\ \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \end{pmatrix} \end{array}$$
- $m_{ii} = 1 \Rightarrow R$ is reflexive
- $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji}$ $\forall i, j$
  $\Rightarrow R$ is antisymmetric

b)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji}$ $\forall i, j$
  $\Rightarrow R$ is antisymmetric

- $M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix} = M$

$\Rightarrow M^{\odot n} = M \ \forall n \in \mathbb{Z} \Rightarrow M^{\odot n} \subseteq M \Rightarrow R$ is transitive.

$\Rightarrow R$ is a partial ordering.

c)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix}$$
(columns 0 1 2 3)

- $\forall i$, $m_{ii} = 1 \Rightarrow R$ is reflexive
- $\forall i \neq j$, $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji}$
  $\Rightarrow R$ is antisymmetric

- $M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix}$

$M^{\odot 2} \not\subseteq M \Rightarrow R$ is not transitive.
we can see that $(3,1), (1,2) \in R$ but $(3,2) \notin R$.

$\Rightarrow R$ is not a partial ordering.

d)
$$\begin{array}{c} \\ 0 \\ 1 \\ 2 \\ 3 \end{array}\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix}$$
(columns 0 1 2 3)

- $\forall i$, $m_{ii} = 1 \Rightarrow R$ is reflexive
- $\forall i \neq j$, $m_{ij} = m_{ji} = 0$, or $m_{ij} \neq m_{ji}$
  $\Rightarrow R$ is antisymmetric

- $M \odot M = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 1 & 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 1 & & & \\ & & & \\ & & & \end{pmatrix} \notin M$

$\Rightarrow R$ is not transitive
since $(1,2), (2,0) \in R$ but $(1,0) \notin R$

e) maximum number of elements of pairs that an antisymetric matrix with diagal elements can have is
$$\frac{n(n-1)}{2} + n = \frac{n(n+1)}{2}$$

for $n=4 \Rightarrow \frac{4(5)}{2} = 10$

however, elements in the brackets are $11 > 10$

Thus, R is not antisymmetric $\Rightarrow$ R is not a partial ordering.

or you can find the matrix and show that it is not antisymmetric. by inspection $(a,1),(1,a) \in R$ with $1 \neq a$ thus, R is not antisymmetric.

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3. Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if

a) a is taller than b?

b) a is not taller than b?

c) $a = b$ or a is an ancestor *of* b?

d) a and b have a common friend?

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a) R is not reflexive $a \not R a$ since $a=a$.
(S,R) is not a poset

b) $a=b$ or a shorter than b. $a \leq b$. This relation is reflexive but not anitsymmetric, since $\exists (a \neq b)$ with a has same height as b.

c) • reflexive : aRa, since $a=a$
• antisymetric : Let $(aRb) \wedge (bRa) \Rightarrow$
1) $a=b$ or a is an ancestor of b
2) $a=b$ or b is an ancestor of a

Since a,b cannot be ancestors of each other $\Rightarrow$ $a=b$
Thus, R is antisymmetric.

• Transitive: Let $a \sim b$, $b \sim c$
$\Rightarrow$ $a=b$ or a is an ancestor of b
$b=c$ or b is an ancestor of c.

proof by cases,

1) suppose $a=b$
   i) if $b \neq c \Rightarrow b$ is an ancestor of c
   $\Rightarrow a$ is an ancestor of $c \Rightarrow a \sim c$
   ii) if $b=c \Rightarrow a=c \Rightarrow a \sim c$

2) suppose a is an ancestor of b
   i) if $b \neq c \Rightarrow b$ is an ancestor of c
   $\Rightarrow a$ is an ancestor of $c \Rightarrow a \sim c$

   ii) if $b=c \Rightarrow a$ is an ancestor of $c \Rightarrow a \sim c$

Thus, $a \sim c \Rightarrow R$ is transitive

$\Rightarrow (S,R)$ is a poset.

d) it's clear that R is not transitive since if
(a,b) have a common friend
(b,c) have a common friend

This does not imply that a and c have a comon friend. These may be two different persons

$\Rightarrow (S,R)$ is not a poset.

4. Is (S, R) a poset if S is the set of all people in the world and $(a, b) \in R$, where a and b are people, if

a) a is no shorter than b?

b) a weighs more than b?

c) $a = b$ or a is a descendant *of* b?

d) a and b do not have a common friend?

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a) $a \geq b$ in height. R is reflexive but not antisymmetric since $\exists (a \neq b)$ with $(a=b)$ in height.

b) Not reflexive since a has same weight as a.

c) This is a poset similar to (3 c)

d) This relation is not reflexive since a has same friends as him/herself.

5. Which of these are posets?

a) $(\mathbf{Z}, =)$ *b*) $(\mathbf{Z}, \neq)$ *c*) $(\mathbf{Z}, \geqslant)$ *d*) $(\mathbf{Z}, \nmid)$

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a) reflexive; $a = a \quad a \in \mathbb{Z}$

antisymmetric : $\forall a, b \in \mathbb{Z}$ if $(a=b \wedge b=a) \rightarrow a=b$

$\Rightarrow R$ is antisymmetric

transitive : $\forall a, b, c \in \mathbb{Z}$ with $a \sim b$, $b \sim c$

$\Rightarrow a=b$, $b=c \Rightarrow a=c \Rightarrow a \sim c$

This is a poset.

b) R is not reflexive, since $a = a$ whenever $a \in \mathbb{Z}$.

c) reflexive : if $a \in \mathbb{Z}$ $a=a \Rightarrow a \geq a \Rightarrow R$ is reflexive

antisymmetry : if $a, b \in \mathbb{Z}$ $(aRb \wedge bRa)$

$\Rightarrow a \geq b$ and $b \geq a \Rightarrow a=b$

$\Rightarrow R$ is antisymmetric

Transitivity : if aRb, $bRc \Rightarrow a \geq b$, $b \geq c$

$\Rightarrow a \geq c \Rightarrow aRc. \Rightarrow R$ is transitive.

This is a poset.

d) R is not reflexive since if $a \neq 0 \in \mathbb{Z}$ $a \nmid a$.

Thus, $a \not R a$.

6. Which of these are posets?

a) $(\mathbb{R}, =)$ *b*) $(\mathbb{R}, <)$ *c*) $(\mathbb{R}, \leqslant)$ *d*) $(\mathbb{R}, \neq)$

a) similar to 5.a This is a poset.

b) R is not reflexive since $a=a$, aka.

c) Similar to 5.c This is a poset

d) Similar to 5.b This is not a poset.

7. Determine whether the relations represented by these zero–one matrices are partial orders.

a) $\begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ c) $\begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

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a) $m_{12} = m_{21} = 1$ Thus, R is not antisymmetric
R is not a partial ordering.

b)
- $\forall i$, $m_{ii} = 1 \Rightarrow$ R is reflexive.
- $\forall i \neq j$, $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji} \Rightarrow$ R is antisymmetric
- $M \odot M = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = M$

Thus, $M^{[2]} = M \subseteq M \Rightarrow$ R is transitive

R is a partial ordering

c)
- $\forall i$, $m_{ii} = 1 \Rightarrow$ R is reflexive
- $\forall i \neq j$, $m_{ij} = m_{ji} = 0$ or $m_{ij} \neq m_{ji} \Rightarrow$ R is antisymmetric
- $M \odot M = \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} \odot \begin{pmatrix} 1 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 1 & 0 \\ & & & \\ & & & \\ & & & \end{pmatrix} \notin M$

$\Rightarrow$ R is not transitive.

This, R is not a partial ordering

8. Determine whether the relations represented by these zero–one matrices are partial orders.

a) $\begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$ b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$ c) $\begin{pmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \end{pmatrix}$

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a) Similar to above R is reflexive and antisymmetric

$$\begin{pmatrix} 1&0&1 \\ 1&1&0 \\ 0&0&1 \end{pmatrix} \odot \begin{pmatrix} 1&0&1 \\ 1&1&0 \\ 0&0&1 \end{pmatrix} = \begin{pmatrix} 1&0&1 \\ 1&1&\textcircled{1} \end{pmatrix} \notin M$$

$\Rightarrow$ R is not transitive

This is not a partial ordering

b) R is reflexive and antisymmetric

$$\begin{pmatrix} 1&0&0 \\ 0&1&0 \\ 1&0&1 \end{pmatrix} \odot \begin{pmatrix} 1&0&0 \\ 0&1&0 \\ 1&0&1 \end{pmatrix} = \begin{pmatrix} 1&0&0 \\ 0&1&0 \\ 1&0&1 \end{pmatrix} = M$$

Thus, $M^{[n]} = M \subseteq M \Rightarrow$ R is transitive

R is a partial ordering

c) R is reflexive and antisymmetric

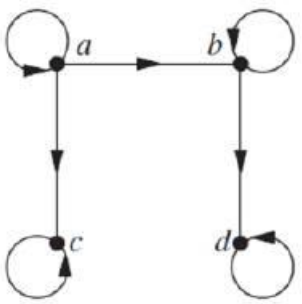
$$\begin{pmatrix} 1&0&1&0 \\ 0&1&1&0 \\ 0&0&1&1 \\ 1&1&0&1 \end{pmatrix} \odot \begin{pmatrix} 1&0&1&0 \\ 0&1&1&0 \\ 0&0&1&1 \\ 1&1&0&1 \end{pmatrix} = \begin{pmatrix} 1&0&1&0 \\ 0&1&1&\textcircled{1} \end{pmatrix} \notin M$$

R is not transitive

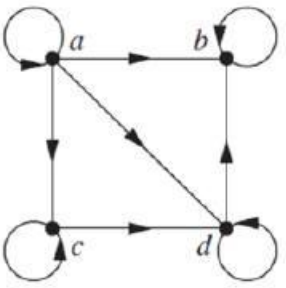
R is not a partial ordering

In Exercises **9–11** determine whether the relation with the directed graph shown is a partial order.

9.



10.



11.



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9) reflexitivity : All loops exist on each vertex
$\Rightarrow$ R is reflexive

antisymmetry: No opposite arrows between vertices exist
$\Rightarrow$ R is antisymmetric.

Transitivity : Triangles between three related vertices does not exist, $(a,b), (b,d) \in R$ but $(a,d) \notin R \Rightarrow$ R is not transitive

This is not a partial order.

10) reflexitivity : loops exist $\Rightarrow$ R is reflexive

antisymmetry : No opposite arrows between vertice exist $\Rightarrow$ R is antisymmetric

transitivity : Not all traiangles between three vertice exist $(c,d), (d,b) \in R$ but $(c,b) \notin R$
$\Rightarrow$ R is not transitive.

This is not a partial order

11) R is reflexive, antisymmetric and transitive.
R is a poset.

12. Let (S, R) be a poset. Show that (S, R^{-1}) is also a poset,where R^{-1} is the inverse of R. The poset (S, R^{-1}) is called the dual of (S, R).

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Suppose (S,R) is a poset

$\Rightarrow$ R is reflexive, anti symmetric and transitive.

$\Rightarrow$ ① aRa
② $\forall a,b \in S \quad (aRb \wedge bRa) \longrightarrow a=b$
③ $\forall a,b,c \in S \quad (aRb \wedge bRc) \longrightarrow aRc.$

Since R^{-1} is the inverse of R. If $(a,b) \in R^{-1}$
$\Rightarrow (b,a) \in R.$
Now, let us check the 3 properties for R^{-1}

① Since $(a,a) \in R \Rightarrow (a,a) \in R^{-1}$
R^{-1} is reflexive

② If $(a,b) \in R \Rightarrow (b,a) \in R^{-1}$
$\Rightarrow (a,b) \notin R^{-1} \Rightarrow$ If $((b,a) \wedge (a,b) \in R^{-1}) \longrightarrow (a=b)$
$\Rightarrow R^{-1}$ is antisymmetric

③ Let $(a,b), (b,c) \in R^{-1}$
$\Rightarrow (b,a), (c,b) \in R$ sine R is transitive
$\Rightarrow (c,a) \in R \Rightarrow (a,c) \in R^{-1}$
$\Rightarrow R^{-1}$ is transitive

$\Rightarrow (S,R^{-1})$ is a poset.

13. Find the duals of these posets.

a) $(\{0, 1, 2\}, \leq)$ *b*) $(\mathbf{Z}, \geq)$

c) $(P(\mathbf{Z}), \supseteq)$ *d*) $\left(\mathbf{Z}^+, |\right)$

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a) $(\{0,1,2\}, \leq)$

$R = \{(0,1), (0,2), (1,2), (0,0), (1,1), (2,2)\}$

$R^{-1} = \{(1,0), (2,0), (2,1), (0,0), (1,1), (2,2)\}$

$R^{-1} \equiv \geq$

$(\{0,1,2\}, \geq)$

b) $(\mathbb{Z}, \leq)$, c) $(P(\mathbb{Z}), \subseteq)$

d) If $a, b \in \mathbb{Z}^+$ $(a,b) \in R$ if $a|b$, $\forall a,b \in \mathbb{Z}^+$ $(b,a) \in R^{-1}$

Then $R^{-1} = \{(a,b) : b|a\}$.

14. Which of these pairs of elements are *comparable* in the poset $\left(\mathbf{Z}^+, |\right)$?

a) 5, 15 *b*) 6, 9 *c*) 8, 16 *d*) 7, 7

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a) $5|15$ $(5,15)$ Thus, they are comparable

b) $6 \nmid 9$ $(6,9) \notin R$ nor $9 \nmid 6$ not comparable.

c) $8|16$ Comparable

d) $7|7$ Comparable

15. Find two *incomparable* elements in these posets.

a) $(P(\{0, 1, 2\}), \subseteq)$ *b*) $(\{1, 2, 4, 6, 8\}, |)$

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a) $P(\{0,1,2\}) = \{\{0\}, \{1\}, \{2\}, \{0,2\} \dots\}$

$\{1\} \not\subseteq \{0,2\}$, $\{1\} \notin \{0,2\}$

and $\{0,2\} \notin \{1\}$

b) 4, 6 $4 \nmid 6$ and $6 \nmid 4$.

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16. Let $S = \{1, 2, 3, 4\}$. With respect to the lexicographic order based on the usual "less than" relation,

a) find all pairs in $S \times S$ less than $(2, 3)$.

b) find all pairs in $S \times S$ greater than $(3, 1)$.

c) draw the Hasse diagram of the poset $(S \times S, \preccurlyeq)$.

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a) So, we can find pair either by $(2,a)$ where $a < 3$ or (b,a) with $b < 2$.

Thus,

$\{(2,1), (2,2), (1,1), (1,2), (1,3), (1,4)\}$

b) $\{(3,2), (3,3), (3,4), (4,1), (4,2), (4,3), (4,4)\}$

c) $S \times S$ contains $4 \times 4 = 16$ points

- This is a total order since each two elements are comparable.
- The hase diagram will begin by $(1,1)$, then $(1,2)$ until $(1,4)$, then $(2,1)$ — $(2,4)$ then, $(3,1)$ — $(3,4)$ then $(4,1)$ — $(4,4)$.

$(4,4)$

|

$(4,3)$

(4,3)

(1,4)

(1,3)

(1,2)

(1,1)

17. Find the lexicographic ordering of these n-tuples:

a) (1, 1, 2), (1, 2, 1) *b*) (0, 1, 2, 3), (0, 1, 3, 2)

c) (1, 0, 1, 0, 1), (0, 1, 1, 1, 0)

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a) $1=1$, $2>1$ $(1,1,2) < (1,2,1)$

b) $(0,1,2,3) < (0,1,3,2)$

c) $(0,1,1,1,0) < (1,0,1,0,1)$

18. Find the lexicographic ordering of these strings of lowercase English letters:

a) quack, quick, quicksilver, quicksand, quacking

b) open, opener, opera, operand, opened

c) zoo, zero, zoom, zoology, zoological

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a) quack < quacking < quick < quicksand < quicksilver

we ordered based on letter appearance.

b) open < opened < opener < opera < operand

c) Zero < Zoo < Zoological < Zoology < Zoom

19. Find the lexicographic ordering of the bit strings 0, 01, 11, 001, 010, 011, 0001, *and* 0101 based on the ordering $0 < 1$.

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we split the set by order of the letter.

$\{0, 01, 11, 001, 010, 011, 0001, 0101\}$

$\{0, 01, 001, 010, 011, 0001, 0101\} < \{11\}$

↑ begins with 0 ↑ begins with 1

↑ order this based on 2nd letter.

$\{0\} < \{001, 0001\} < \{010, 011, 0101\} < \{11\}$

↑ last number of 0's ↑ 2nd letter 0 ↑ 2nd letter 1

⇓

$\{0\} < \{0001\} < \{001\} < 01 < \{010, 011, 0101\} < \{11\}$

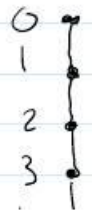
↑ third letter 0 ↑ third letter 1 ↑ shortest

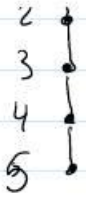
$0 < 0001 < 001 < 01 < 010 < 0101 < 011 < 11$

20. Draw the Hasse diagram for the "greater than or equal to" relation on $\{0, 1, 2, 3, 4, 5\}$.

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This is a Total order Thus hasse diagram looks like

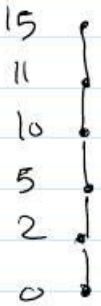




21. Draw the Hasse diagram for the "less than or equal to" relation on $\{0, 2, 5, 10, 11, 15\}$.

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This is also a total order. Since has diagram removes reflexive and transitivity edges we just need to order them in an ascending manner.

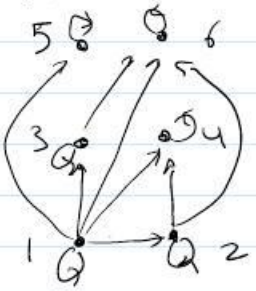


22. Draw the Hasse diagram for divisibility on the set

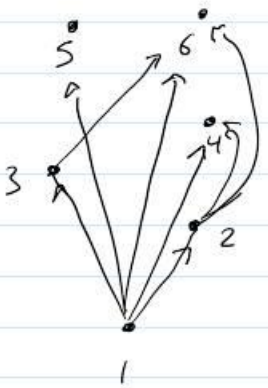
a) $\{1, 2, 3, 4, 5, 6\}$.
b) $\{3, 5, 7, 11, 13, 16, 17\}$.
c) $\{2, 3, 5, 10, 11, 15, 25\}$.
d) $\{1, 3, 9, 27, 81, 243\}$.

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a) First draw R.



remove loops



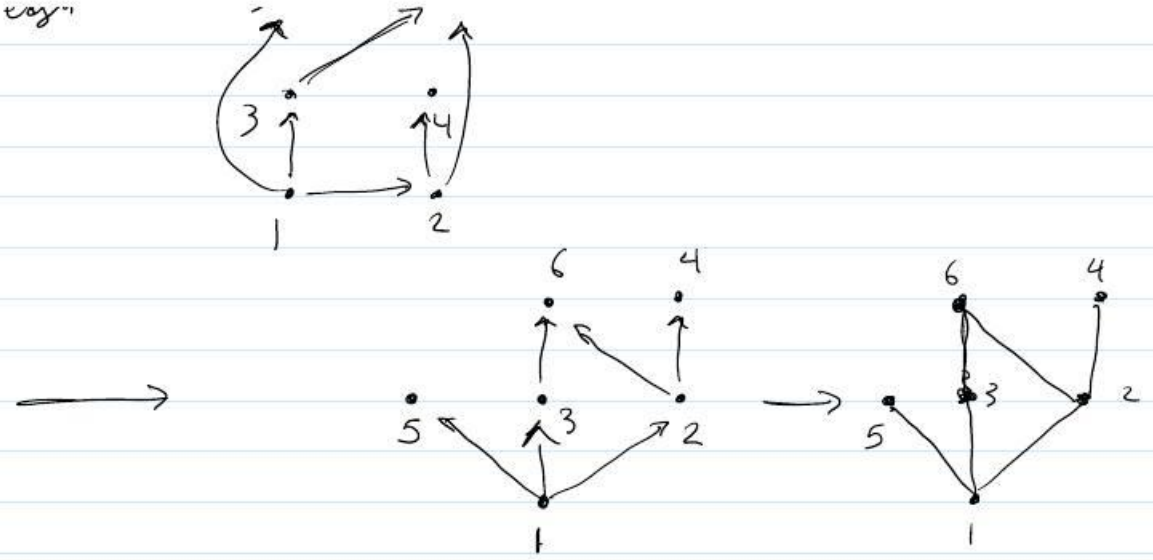
(1,2), (2,4)
~~(1,4)~~

(1,3) (3,6)
~~(1,6)~~

remove transitive edges



eg:

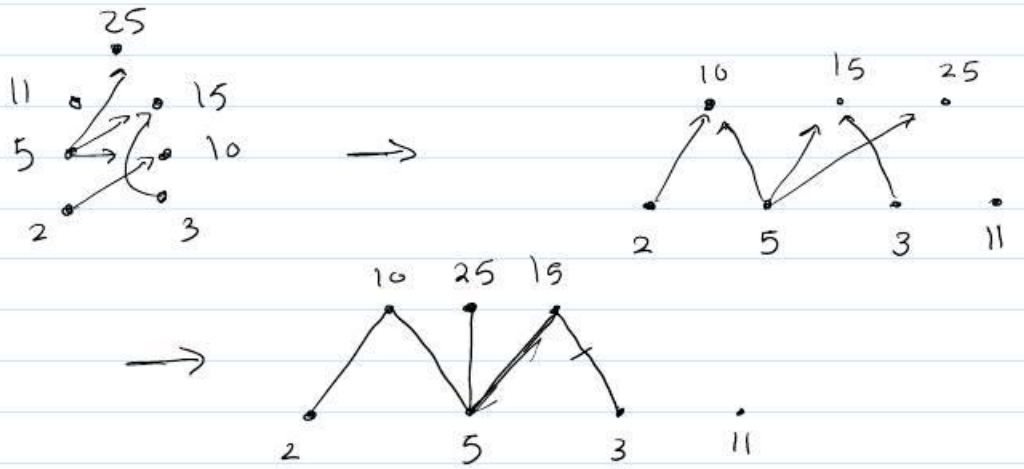


b) all numbers are primes except $16 = 2 \times 8 = 2^4$
This, all numbers are relatively prime $\Rightarrow$ no connector



c) Write R without loops

$\{ (2,10), (3,15), (5,10), (5,15), (5,25) \}$



d) $\{1, 3, 9, 27, 81, 243\}$

R without loops

R without loops

{ (1,3), (1,9), (1,27), (1,81), (1,243)
(3,9) (3,27) (3,81), (3,243)
(9,27), (9,81), (9,243)
(27,81), (27,243)
(81,243) }

↓

find transitivity between 1st & 2nd row.

(1,3), (3,9) → (1,9)
(1,3), (3,27) → (1,27)
(1,3), (3,81) → (1,81)
(1,3), (3,243) → (1,243)

} remove

↓

H = { (1,3), (3,9), (3,27), (3,81), (3,243)
(9,27), (9,81), (9,243)
(27,81), (27,243)
(81,243) }

find between 1st & 2nd row

(3,9), (9,27) → (3,27)
(3,9), (9,81) → (3,81)
(3,9), (9,243) → (3,243)

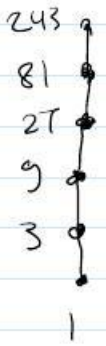
} remove

H = { (1,3), (3,9), (9,27), (9,81), (9,243)
(27,81), (27,243)

$(27, 81)$, $(27, 243)$
$(81, 243)$ }

$(9, 27)$, $(27, 81) \longrightarrow (9, 81)$
$(9, 27)$, $(27, 243) \longrightarrow (9, 243)$
$(27, 81)$, $(81, 243) \longrightarrow (27, 243)$
} remove

$H = \{ (1,3), (3,9), (9,27), (27,81), (81,243) \}$



24. Draw the Hasse diagram for inclusion on the set $P(S)$, where $S = \{a, b, c, d\}$.

$P(S) = \{ \{a\}, \{b\}, \{c\}, \{d\}$
$\{a,b\}, \{a,c\}, \{a,d\}$
$\{b,c\}, \{b,d\}, \{c,d\}.$
$\{a,b,c\}, \{a,b,d\}, \{a,c,d\}, \{b,c,d\}$ }
$\{a,b,c,d\}$ }

R, $\subseteq$ we will denote set as $A_{ab} = \{a,b\}$ and so, on . first cardor $A_{a...}$ relation

$\{ (A_a, A_{ab}), (A_a, A_{ac}), (A_a, A_{ad})$

$(A_a, A_{abc}), (A_a, A_{abd}), (A_a, A_{acd})$

(A_a, A_{abcd})

$(A_{ab}, A_{abc}), (A_{ab}, A_{abd}), (A_{ac}, A_{abc})$

$(A_{ac}, A_{acd}), (A_{ad}, A_{abd}), (A_{ad}, A_{acd})$

$(A_{ab}, A_{abcd}) \}$ ← This is a small subset.

remove transitivity edges

$(A_a, A_{ab}), (A_{ab}, A_{abc}) \rightarrow (A_a, A_{abc})$ ← remove

$\Rightarrow$ subset with one character will only connect to two characters unless. "claim"

Notice that this is in general for any $x, y, z = \{a, b, c, d\}$

$$(A_x, A_{xy}), (A_{xy}, A_{xyz}) \rightarrow (A_x, A_{xyz})$$

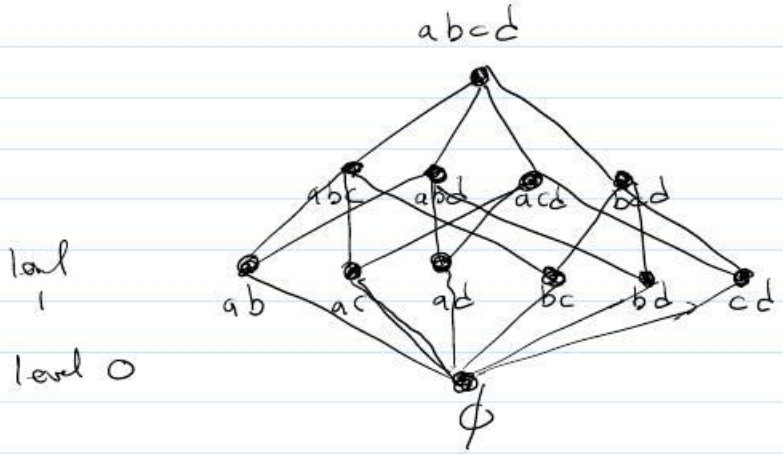
Thus, level one connects only to two. similarly we can show that level one doesn't connect to level 4 $\{a, b, c, d\}$. In the same way

$$(A_{ab}, A_{abc}), (A_{abc}, A_{ab,cd}) \in R$$

$$\rightarrow (A_{ab}, A_{abe,d})$$

Thus, level 2 doesn't connect to level 4.

Now, we can draw haze diagram begging from ϕ set.

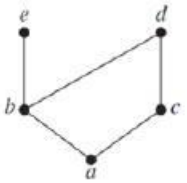


In Exercises **25–27** list all ordered pairs in the partial ordering with the accompanying Hasse diagram.

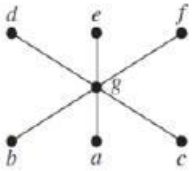
25.



26.

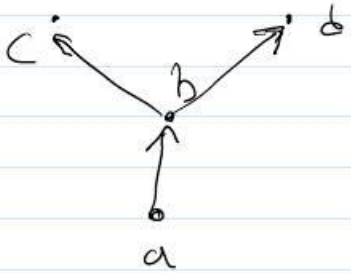


27.

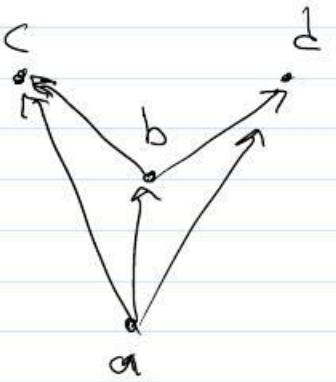


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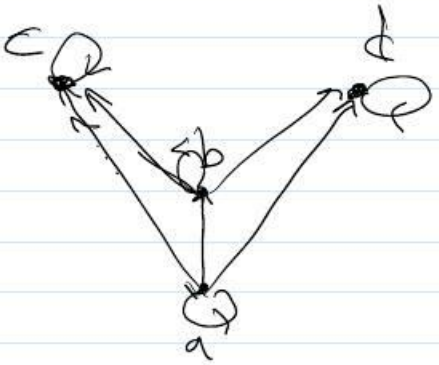
25) So, we just need to reconstruct the diagram by forming arrow then adding transitive relation then reflexivity



Transitivity



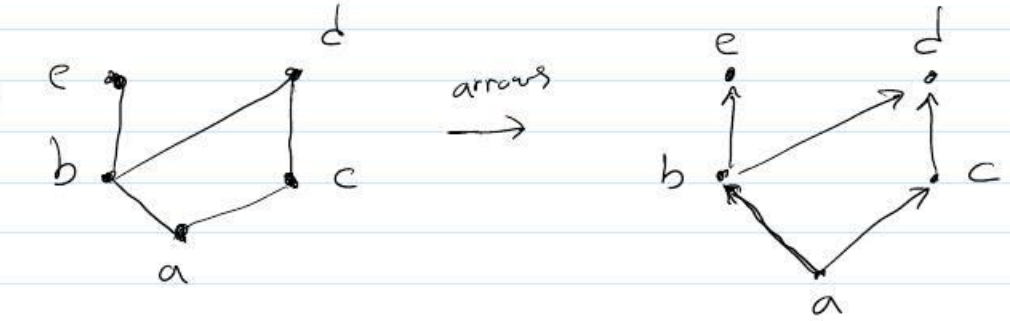
reflexivity

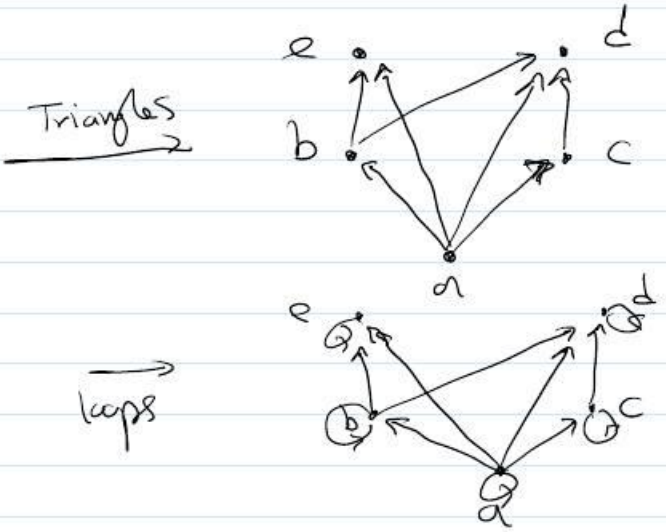


$R = \{ (a,a), (c,c), (b,b), (d,d)$

$R = \{ (a,a), (c,c), (b,b), (d,d)$
$(a,b), (b,d), (a,d), (a,b), (b,c), (a,c) \}$

26)

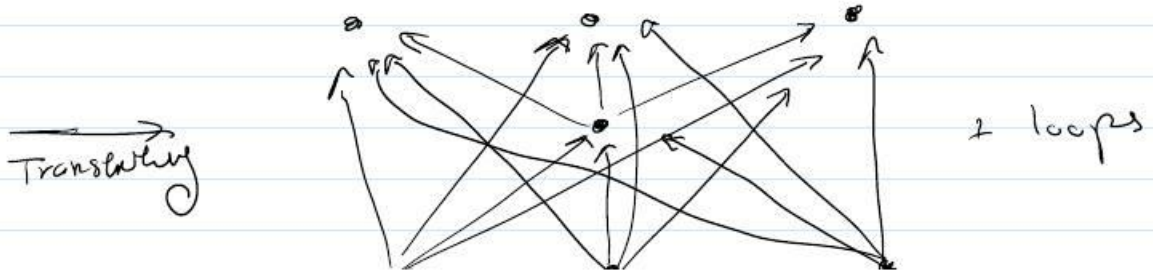


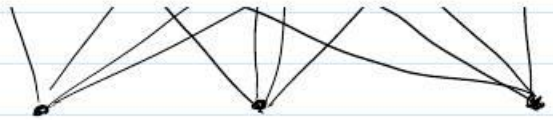


$R = \{ (a,a), (b,b), (c,c), (d,d), (e,e)$
$(a,b), (b,e), (a,e), (a,c), (c,d), (a,d)$
$(b,d), \}^r$

27)







$R = \{ (a,a), (b,b), (c,c), (d,d), (e,e), (f,f), (g,g)$
$(b,g), (g,d), (b,d), \{$ and so on.. $\}$